Solids, Nets, Surface Area and Volume

Concept

Surface area measures the outside covering; volume measures space inside.

Right prism

$V = Bh$; $SA = 2B + Ph$. B = base area, P = base perimeter.

Cylinder

$V = \pi r^2 h$; closed $SA = 2\pi r^2 + 2\pi rh$.

Nets and views

Unfold all faces; top/front/side views may hide cubes.

Units

Area uses square units; volume uses cubic units. Remove missing faces.

Example 1

Find volume and surface area of a $6 \times 4 \times 3$ cm box.

Solution

1. $V = 6 \times 4 \times 3 = 72\ \text{cm}^3$.

2. $SA = 2(6\times4 + 6\times3 + 4\times3) = 108\ \text{cm}^2$.

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Example 2

A closed cylinder has radius 3 cm and height 8 cm. Find V and SA.

Solution

1. $V = \pi(3^2)(8) = 72\pi \text{ cm}^3 \approx 226.19 \text{ cm}^3$.

2. $SA = 2\pi(3^2) + 2\pi(3)(8) = 66\pi \text{ cm}^2 \approx 207.35 \text{ cm}^2$.

Example 3

A triangular prism has a 3-4-5 right-triangle base and length 10 cm.

Solution

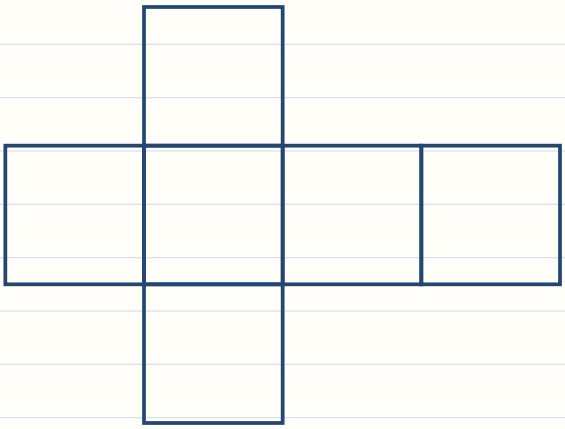
1. Base area $B = (3\times4)/2 = 6 \text{ cm}^2$; perimeter $P = 3+4+5 = 12$ cm.

2. $V = 6(10) = 60 \text{ cm}^3$; $SA = 2(6) + 12(10) = 132 \text{ cm}^2$.

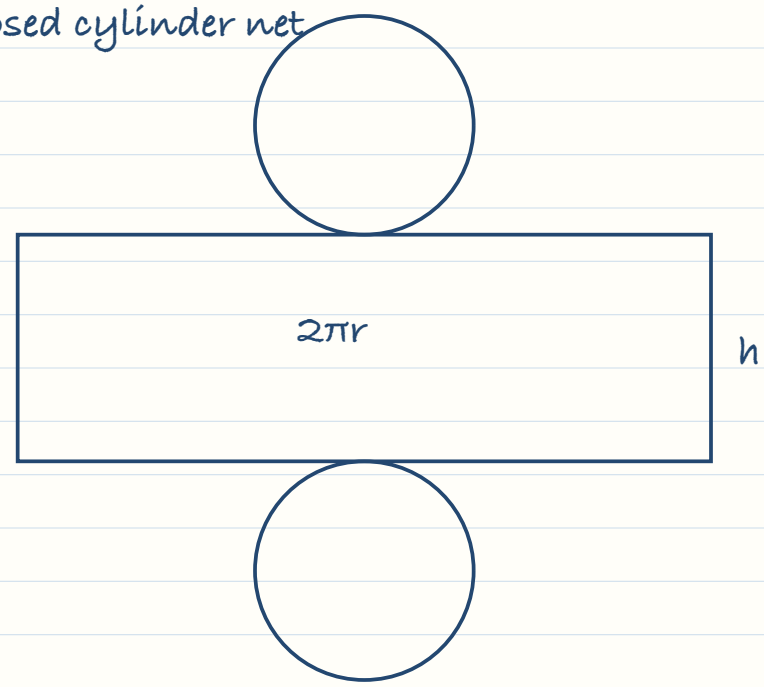
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Visual notes | nets

A cube net: six equal squares



A closed cylinder net



Cylinder: two circles and a 2πr by h rectangle.

Sketches are schematic; match shared edges when folding.

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Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Find V and SA of a cube with side 5 cm.

2. Find V of a cylinder with diameter 10 cm and height 6 cm.

3. An open box has length 8, width 5 and height 2 cm. Find its surface area.

4. What rectangle is needed for the curved surface of a cylinder of radius 2 cm and height 7 cm?

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Practice answers | check after trying

1. $V = 125\ cm^3$; $SA = 6 \times 25 = 150\ cm^2$.

2. Radius $= 5$, so $V = 150\pi\ cm^3 \approx 471.24\ cm^3$.

3. Base + four walls $= 40 + 2(16) + 2(10) = 92\ cm^2$.

4. A rectangle 4π cm by 7 cm. Its length wraps exactly once around the circle.